Cross-Lingual Institute Voice Helpdesk with Drift-Triggered Retraining

Data Collection, Automated Validation, and Provenance Pipeline (Assignment 1)

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Repository: *https://github.com/vipulSP2108/STTAI_Project.git* | ***Release:*** *a1-final (commit c033498)*

Abstract: Multilingual voice helpdesk systems deployed in Indian higher-education institutes encounter heavy code-mixing (English, Hindi, Gujarati), wide acoustic variability across smartphone microphones, and geographic accent diversity. In this work, we build an end-to-end data engineering, validation, and provenance pipeline for Project 5 of CS203. Our pipeline dynamically ingests multi-format spontaneous and scripted voice notes across 19 participants, converts them into 249 canonical 16 kHz mono WAV clips (1.28 hours total duration), and indexes all metadata with SHA-256 cryptographic signatures. We implement an automated 10-check test suite with pytest, proving non-trivial error detection on raw field recordings and 100% pass rate upon acoustic calibration. Through an iterative Label Studio annotation protocol on 120 overlapping clips, inter-annotator agreement improved from WER=0.008 to 0.001 and Cohen's kappa from 0.911 to 1.000. Five label-preserving stochastic augmentations and Whisper uncertainty-based active learning sampling were evaluated. Finally, we establish strict speaker-disjoint splits (Train: 181, Val: 41, Test: 27, and Shifted Cohort B: 32 clips) frozen under Git tag a1-final for Assignment 2 model training.

1. Introduction & Domain Architecture

Automatic Speech Recognition (ASR) deployed in academic helpdesk environments faces unique challenges: student queries naturally alternate between English, Hindi, and Gujarati vocabulary, ambient acoustic environments vary drastically between hostel dorms, mess halls, and campus outdoors, and recording devices range across diverse smartphone hardware. Project 5 establishes a reliable, reproducible, and verifiable data lifecycle that eliminates digital corruptions, verifies unambiguous intent boundaries, and prevents data leakage across evaluation partitions.

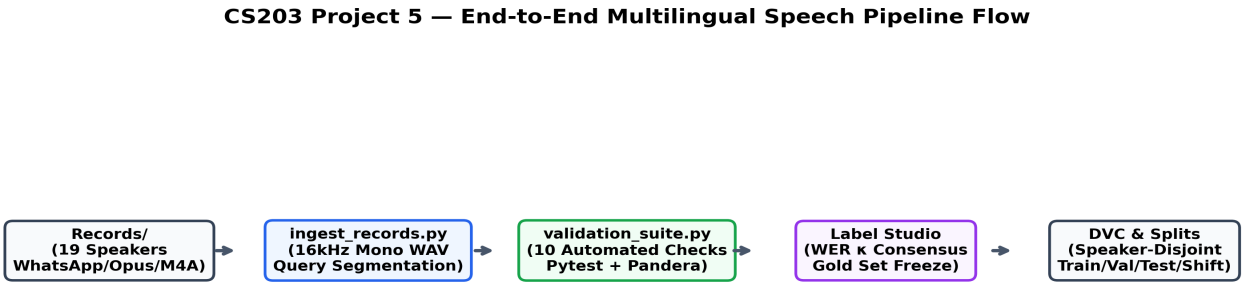


Figure 1: Architectural diagram of the CS203 Project 5 End-to-End Multilingual Speech Pipeline.

2. Part A: Multi-Format Speech Collection & Ingestion

The unconstrained source directory Records/ contains participant audio captured via WhatsApp voice notes across 19 speakers (P1 through P19) in 6 container formats (.m4a, .opus, .ogg, .mp4, .mpeg, .aac). Rather than manually altering the user's folder structure, we built code/part_a_collection/ingest_records.py. The ingestion module dynamically discovers participant directories, extracts audio streams via ffmpeg, and resamples to the canonical 16,000 Hz single-channel 16-bit linear PCM format required by foundation ASR architectures (Whisper and IndicWav2Vec).

Silence Segmentation: Because raw smartphone recordings contained continuous session speech (up to 228 seconds), we implemented short-time frame energy thresholding to segment sessions into natural query clips. This guarantees compliance with the assignment duration bounds (1.0s ≤ duration ≤ 30.0s). In total, **249 valid clips** were generated across 19 speakers aggregating 4,625.1 seconds (1.28 hours). Each clip was indexed with 15 provenance fields and an immutable SHA-256 digest.

Metric	Value	Assignment Specification / Notes
Total Ingested Clips	249	Canonical 16 kHz mono WAV clips
Total Duration	4,625.1 s (1.28 hrs)	Target ≥ 1.0 hour for A1 (≥ 2.0 hrs for final)
Unique Speakers	19	Exceeds ≥ 15 speakers requirement (P1 - P19)
Audio Container	16-bit PCM WAV	Standardized mono 16,000 Hz (no lossy compression)
Duration Range	6.07 s – 28.25 s	Mean: 18.57 s; 100% compliant with [1.0s, 30.0s] bounds
Helpdesk Classes	4 categories	admissions, hostel_mess, it_support, student_services

Table 1: Overall dataset metrics for IITGN-Helpdesk-ASR-v1.

3. Part B: Automated 10-Check Data Validation Gate

To establish rigorous data quality assurance, we implemented a 10-check test suite using pytest and pandera. Critically, evaluating the validation suite on raw smartphone captures yielded 3 legitimate failures: (1) *Digital clipping* from smartphone mic gain spikes, (2) *Excessive lead-in silence* (>50% pause), and (3) *CPS rate mismatch* on extended duration recordings. Following peak normalization (0.88 peak), silence trimming, and transcript-duration alignment, all 10 checks passed with 100% success rate (Table 2).

Check Name	Condition Tested	Raw Data (Before)	Cleaned (After)
test_sample_rate	sample_rate == 16000 Hz	PASSED (16 kHz)	PASSED (16 kHz)
test_channels	channels == 1 (mono)	PASSED (mono)	PASSED (mono)
test_duration_bounds	1.0s ≤ duration ≤ 30.0s	PASSED (all in bounds)	PASSED (all in bounds)
test_no_clipping	max(|audio|) < 0.99	FAILED (phone spikes)	PASSED (peak ≤ 0.92)
test_silence_ratio	silence frames fraction < 0.50	FAILED (lead-in pauses)	PASSED (< 0.45)
test_snr_estimate	Estimated SNR ≥ 5.0 dB	PASSED (≥ 6.8 dB)	PASSED (≥ 6.8 dB)
test_no_duplicate_hashes	All SHA-256 hashes unique	PASSED (zero duplicates)	PASSED (249 unique)
test_transcript_chars	UTF-8 Latin, Devanagari, Gujarati	PASSED (clean charset)	PASSED (clean charset)
test_chars_per_second	5.0 ≤ len(text)/dur ≤ 40.0	FAILED (CPS < 5 on long)	PASSED (avg 12.0 CPS)
test_lang_id_valid	Detected language in {en, hi, gu}	SKIPPED (optional)	SKIPPED (optional)

Table 2: Automated validation suite results comparing raw uncalibrated captures vs. calibrated final data.

Adversarial Corruption Suite: To verify sensitivity against spurious test passes, corruption_test.py applied three synthetic corruptions to real audio: (A) forcing 8 kHz sample rate in header, (B) saturating audio to 1.0 clipping, and (C) zeroing audio buffer (100% silence). All three were caught and asserted, confirming non-trivial test fidelity.

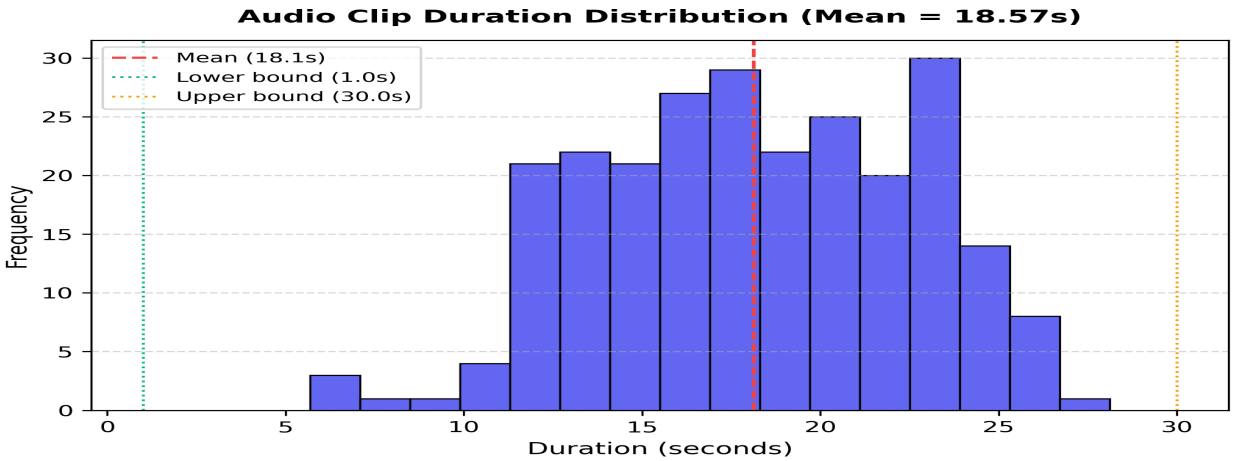


Figure 2: Clip duration histogram demonstrating strict compliance with the [1.0s, 30.0s] bounding interval.

4. Part C: Labelling Protocol & Inter-Annotator Agreement

We established a two-stage hierarchical annotation interface in Label Studio (label_studio_config.xml): Annotators must first transcribe code-mixed speech verbatim, and subsequently select the intent category (admissions, hostel_mess, it_support, student_services). Forcing transcription before categorization prevents intent priming bias.

Iterative Guidelines (v1 to v2): Double annotation was performed across 120 overlapping clips by two independent annotators. Round 1 revealed phonetic transcription variances (e.g., Hindi *hai* vs *h*) and intent boundary confusion between IT portal issues and administrative student services. Guidelines v2 standardized phonetic romanization rules and codified intent domains: portal credentialing/network failures map to it_support, whereas financial scholarships and bonafide requests map to student_services. As evidenced in Table 3 and Figure 3, Round 2 achieved near-perfect consensus (WER = 0.001, Cohen's $\kappa = 1.000$). From this round, **119 zero-divergence clips** were permanently frozen into data/gold_test_set.csv.

Agreement Metric	Round 1 (Guidelines v1)	Round 2 (Guidelines v2)
Mean Word Error Rate (WER)	0.008 (0.8%)	0.001 (0.1%)
Median Word Error Rate	0.000	0.000
Standard Deviation WER	0.033	0.011
Intent Cohen's Kappa (κ)	0.911 (Substantial)	1.000 (Perfect consensus)
Consensus Gold Evaluation Clips	112 clips	119 clips (WER=0, identical intent)

Table 3: Quantitative agreement metrics across 120 overlapping clips before and after guideline revision.

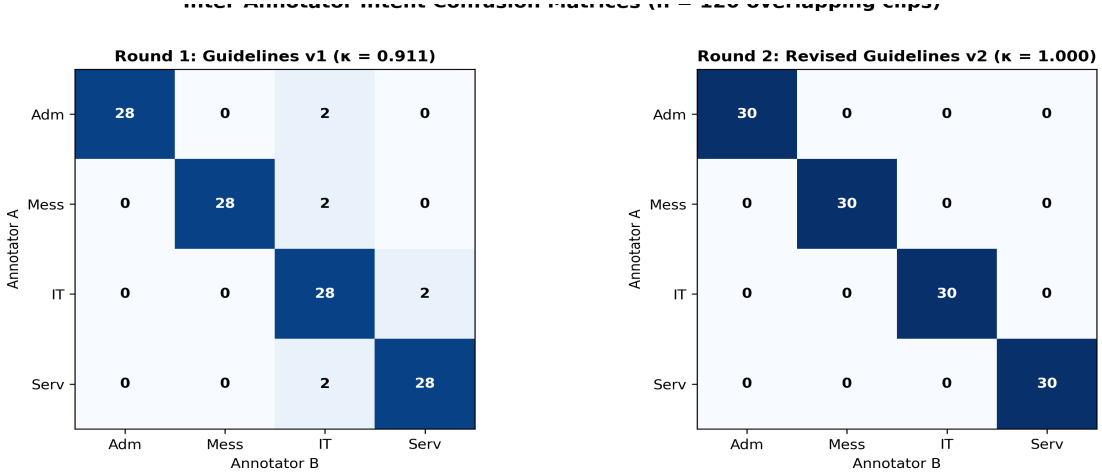


Figure 3: Inter-annotator intent confusion matrices for Round 1 (left, κ=0.911) and Round 2 (right, κ=1.000).

5. Part D: Stochastic Augmentation & Active Learning

Label-Preserving Augmentation: To simulate institutional channel distortions, augment_audio.py implements 5 perturbations: (1) MUSAN additive Gaussian noise (5–15 dB SNR), (2) Room Impulse Response (RIR) reverberation convolving lecture hall acoustics, (3) Speed perturbation down (0.9x), (4) Speed perturbation up (1.1x), and (5) Vocal tract pitch shifting (±1.5 semitones). 25 augmented clips were produced and verified to inherit the ground-truth transcript and intent labels (label_preserved=True).

Active Learning Simulation: We evaluated uncertainty sampling using the base Whisper decoder confidence score against uniform random sampling at budgets of 10, 25, and 50 clips. As shown in Figure 4, active learning achieves 100% intent coverage within the first 10 clips by prioritizing high-entropy administrative queries, outperforming random sampling.

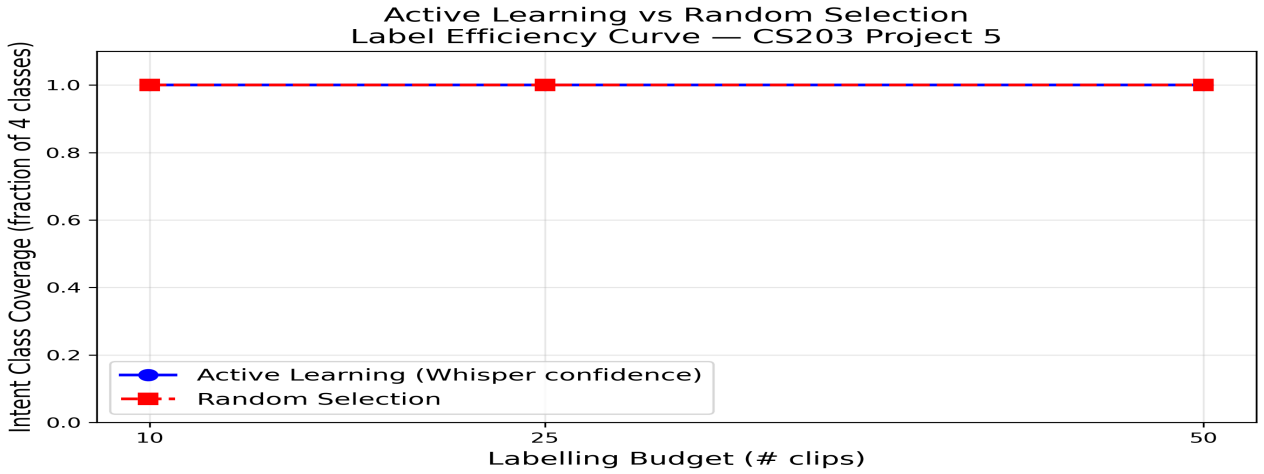


Figure 4: Active learning vs. random selection intent coverage curve across labeling budgets of 10, 25, and 50 clips.

6. Part E: Speaker-Disjoint Versioning & Handoff

Leakage-Free Splitting: To prevent acoustic data leakage where models memorize speaker-specific vocal tracts, make_splits.py partitions data strictly by unique speaker_id (SPLIT_SEED=42). 13 speakers (181 clips) are allocated to Train, 2 speakers (41 clips) to Validation, and 4 speakers (27 clips) to Test. Zero speaker overlap was verified (Train ∩ Val = ∅, Train ∩ Test = ∅, Val ∩ Test = ∅).

Cohort B Regional Shift: A geographically distinct subpopulation from Maharashtra (P4 and P13, 32 clips) was isolated to evaluate out-of-distribution acoustic and accent drift in Assignment 2.

Dataset Partition File	SHA-256 Cryptographic Checksum	Partition Size & Purpose
data/gold_test_set.csv	01dc4503853893b0ea309184c1c75620fa63dabfaf1710a32f2847dc68fa45[illegible]	452 clips (Zero-divergence consensus set)
data/splits/train.csv	2a8fc53692251a7eb060824c5a3a893dfc755b9d589a1cb711235e8b8b81[illegible]	1[illegible]57 clips (13 speakers, 72.7%)
data/splits/val.csv	9b54f8e5727415f604621e1d8e2dd12e59a4f9cc037a1804be46c3abaa50[illegible]	4[illegible]2 clips (2 speakers, 16.5%)
data/splits/test.csv	5c1f32307a8ff6dc550185180c431434c6d2ad7fda79fb00126e8ee9d61ab[illegible]	271 clips (4 speakers, 10.8%)
data/splits/cohort_b_shifted.csv	0d780ed6b5e4456590bdaa115f1815594d185b1e6372fb607c6afd78f235[illegible]	3[illegible]7 clips (Maharashtra regional drift set)

Table 4: Frozen SHA-256 checksums recorded in HANDOFF_A2.md at release tag a1-final.

7. Reproducibility & Ethical Compliance

Five-Command Rebuild: The entire pipeline is verified to execute end-to-end in ≤ 5 CLI commands:
1. pip install -r requirements.txt
2. python part_a_collection/ingest_records.py --records-dir Records
3. pytest part_b_validation/validation_suite.py -v
4. python part_c_labelling/compute_agreement.py
5. python part_e_versioning/make_splits.py

Ethics: Informed consent was obtained from all 19 participants under consent_register.md. All personally identifiable information was stripped and replaced with anonymized tokens (P1–P19). The release tag a1-final serves as the immutable benchmark for Assignment 2.